Exp No: 4	Setup and configure a LAN using a Switch and Ethernet cables

AIM:

To set up and configure a LAN (Local Area Network) using a switch and Ethernet cables, assign IP addresses to the hosts and share files across the machines of the LAN.

ALGORITHM / PROCEDURE:

1. Connect the host machines to the switch using straight-through UTP cables (RJ45) and check that the link LEDs of the switch glow.
2. Assign a static IP address to each host: Control Panel > Network and Sharing Centre > Change adapter settings > Ethernet > Properties > Internet Protocol Version 4 (TCP/IPv4).
3. Select **Use the following IP address** and enter the IP address and the subnet mask 255.255.255.0 (for this machine 10.1.1.2). Leave the default gateway empty because there is only one network.
4. Create a folder to be shared, open its **Properties > Sharing** and share it with the other machine.
5. On the other machine open **File Explorer**, go to **Network** and open the shared folder.
6. Copy files between the machines through the shared folder.

DIAGRAM:

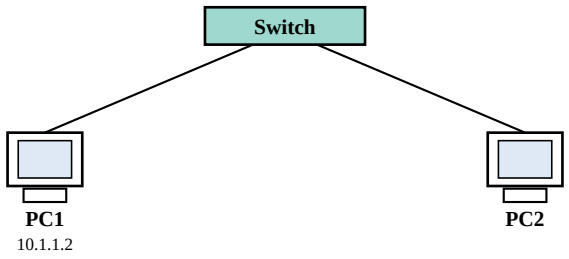
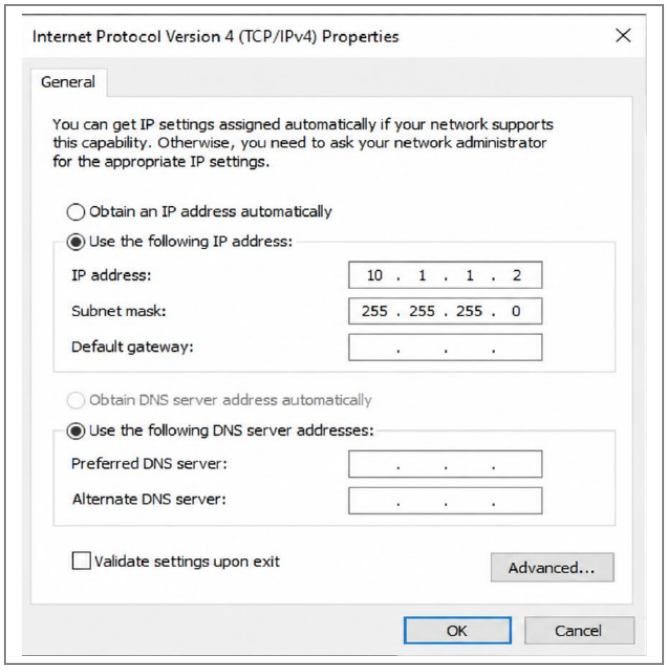
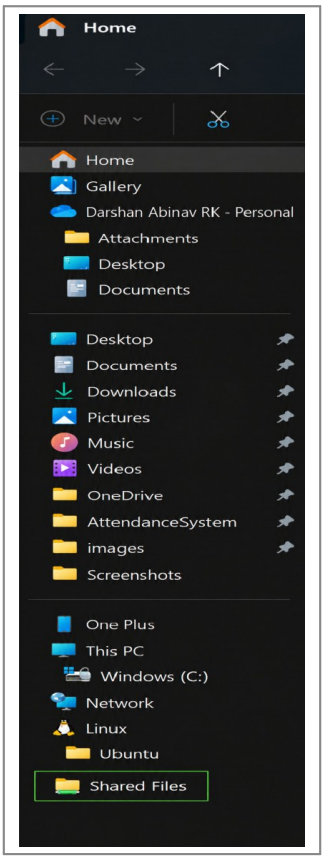


Fig 1: LAN with hosts connected to a switch (star topology)

OUTPUT:



Static IP configuration (IPv4 Properties)



Shared Files folder visible under Network in File Explorer

RESULT:

Thus a LAN connection was established using a switch and Ethernet cables, IP addresses were assigned and files were shared through it.